

Automated Network Request Management in ServiceNow

Project Report

Date	09 September 2026
Student Name	Vijjapu Madhuri
Project Name	Automated Network Request Management in ServiceNow

1. INTRODUCTION

1.1 Project Overview

1.2 Purpose

2. IDEATION PHASE

2.1 Problem Statement

2.2 Empathy Map Canvas

2.3 Brainstorming

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

3.2 Solution Requirement

3.3 Data Flow Diagram

3.4 Technology Stack

4. PROJECT DESIGN

4.1 Problem Solution Fit

4.2 Proposed Solution

4.3 Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

7. RESULTS

7.1 Output Screenshots

8. ADVANTAGES & DISADVANTAGES

9. CONCLUSION

10. FUTURE SCOPE

11. APPENDIX

Source Code(if any)

Dataset Link

GitHub & Project Demo Link

1. INTRODUCTION

1.1 Project Overview

The Automated Network Request Management in ServiceNow project is designed to provide a centralized platform for submitting, processing, approving, tracking, and managing network-related service requests. The system uses the ServiceNow platform to automate the complete request lifecycle and reduce manual effort.

The application provides a Network Request Catalog Item through the ServiceNow Service Portal. Users can enter the required network request information, submit the request, and track its progress. The submitted request is processed through an automated workflow using Flow Designer, where approval, record creation, notifications, and status updates are handled systematically.

The project improves the efficiency of network service management by maintaining request information in a centralized Network Database and providing controlled access through ServiceNow security and ACL configurations.

1.2 Purpose

The main purpose of this project is to automate the network request management process using ServiceNow.

The project aims to:

- Simplify the process of submitting network-related requests.
- Reduce manual processing and administrative effort.
- Automate request validation and workflow processing.
- Route requests for approval automatically.
- Store request information in a centralized Network Database.
- Send automated email notifications for important request updates.
- Allow users to track request and approval status.
- Provide controlled access to network request records.
- Improve transparency and efficiency throughout the request lifecycle.

2. IDEATION PHASE

2.1 Problem Statement

In a manual network request management process, users may need to communicate their requirements through emails, messages, or other channels. This can result in delays, incomplete information, difficulty in tracking requests, and additional manual work for administrators.

There may also be no centralized location to maintain all network request information, making it difficult to monitor request progress and approval status.

The proposed project addresses these issues by providing a centralized ServiceNow-based system where users can submit network requests through a structured form. The system automates processing, approval, notification, and status tracking.

Problem Statement:

"To develop an automated and centralized network request management system using ServiceNow that simplifies request submission, automates approval and processing, provides real-time status visibility, and reduces manual intervention."

2.2 Empathy Map Canvas

Empathy Map Canvas

Dimension	Employee / Network Requester Perspective
Says	Needs an easy way to raise a network request; expects the required details to be clear and the request to move without unnecessary delays.
Thinks	Checks the ServiceNow portal, opens the network request option, enters connection and contact details, and submits the form.
Does	Expects the request to be handled smoothly and feels more confident when the current status is easy to follow.
Feels	Repeated data entry, missing details, unclear next steps, limited progress updates, and delays in fulfilment.

Pain Points	Guided request entry, accurate information capture, timely processing, visible progress, and dependable service delivery.
Needs / Goals	Simple and intuitive request submission, fast turnaround, status visibility, faster request fulfilment, and reliable service delivery.

End User (Requester) — Empathy Map

“I want to raise a network request without confusion or repeated follow-up.” “I want a simple and intuitive way to submit a network service request.”		Expects the form to ask for the right details and make the next step clear. Wants the request form to clearly capture the required details and expects a straightforward process.
	REQUESTER	
Opens the Service Portal, selects the network request option, enters the required information, and submits it. Accesses the Service Portal, searches for Network Requests, fills in the required details, and submits the request.		Feels more comfortable when the request progress and approval status are visible. Wants a faster, more reliable experience; may feel uncertain when status visibility is limited.

2.3 Brainstorming

Idea Prioritization

The prioritization below considers the main building blocks of the Automated Network Request Management project — request entry, form data, workflow automation, approvals, records, communication, security, and validation. Each item is rated High or Medium according to its contribution to a smoother request experience, implementation practicality, and dependency on other features.

Idea Group / Feature	Purpose in Project	Priority	Reason for Priority
Network Request Service Catalog	Service Catalog item (Category: Network) provides a standard starting point for employees to raise new or existing connection requests.	High	Creates a consistent entry point and avoids requests being handled through disconnected manual channels.
Network Request Form and Variables	Collects details such as Requested For, Mobile Number, Type of Connection, Existing ID, Total Amount, Mode of Payment, Address, and the Opened on Behalf Of variable set.	High	Improves the quality of information received by the fulfilment team and reduces back-and-forth with the requester.
Flow Designer Automation	Uses Get Catalog Variables, Create Record, Send Email, Ask for Approval, If/Else logic, and Update Record to move the request through defined stages after submission.	High	Connects the major steps into one automated lifecycle and reduces dependency on manual coordination.
Approval Routing	Uses the Ask for Approval action and Approval Request related list so requests can be reviewed before fulfilment.	High	Introduces a controlled checkpoint and makes the approval decision traceable.
Network Database Record Creation	Creates a Network Database record containing the information needed by administrators and the network fulfilment team to work on the request.	High	Provides one operational record for assignment, processing, approval information, and follow-up.
Dynamic Form Behaviour	Uses a Catalog UI Policy so the Existing ID field is shown when the requester selects an existing connection.	Medium	Keeps the form relevant to the selected connection type and avoids asking for unnecessary information.
Email Notifications	Uses the Send Email action to inform relevant users when important request events occur.	Medium	Keeps stakeholders informed without requiring them to repeatedly check the system.
Request Tracking / Status Visibility	Uses the Approval Status value on the Network Database record to indicate the current position of the request.	Medium	Helps users understand whether a request is awaiting action, approved, or requires attention.

Idea Group / Feature	Purpose in Project	Priority	Reason for Priority
Access Control	Applies ACLs and Access Control Rules to limit who can view or update sensitive Network Database information.	Medium	Supports role-based access and protects request information while depending on the underlying table design.
Validation and Testing	Checks mandatory data, form behaviour, record creation, notification steps, and approval outcomes through testing and User Acceptance Testing.	Medium	Helps identify workflow or data issues before the solution is used in the live environment.

Prioritization Overview

The card view below summarizes the main project components together with their assigned priority levels.

Network Request Service Catalog	High
Network Request Form and Variables	High
Flow Designer Automation	High
Approval Routing	High
Network Database Record Creation	High
Dynamic Form Behaviour	Medium
Email Notifications	Medium
Request Tracking / Status Visibility	Medium
Access Control	Medium
Validation and Testing	Medium

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey represents the steps followed by a user from submitting a network request until the request is processed.

Stage	User Activity	System Response	User Experience
1. Request Identification	User identifies the need for a network connection/service.	User accesses the ServiceNow Service Portal.	Easy access to the service.

2. Request Submission	User opens the Network Request Catalog Item.	System displays the request form.	Simple and structured form.
3. Information Entry	User enters requester, connection, payment, amount, and address details.	System validates the entered information.	Guided data entry.
4. Conditional Selection	User selects the type of connection.	Existing ID is displayed when Existing connection is selected.	Relevant fields are shown dynamically.
5. Request Creation	User submits the request.	Flow Designer processes the request and creates the required record.	Request is processed automatically.
6. Approval	Request moves through the approval process.	Approval request is generated.	Clear approval process.
7. Notification	Request progresses through workflow stages.	Automated email notifications are generated.	User receives updates.
8. Status Tracking	User checks the request progress.	Request/Approval Status is updated.	Better visibility.
9. Completion	Request completes the required process.	Final status is reflected in the system.	Complete and traceable process.

3.2 Solution Requirement

Functional Requirements

Requirements are grouped in the order they execute during a request's lifecycle, from intake through verification.

ID	Capability	What It Covers	Depends On
FR-01	Network Request Submission	The system shall provide a Network Request Catalog Item through the ServiceNow Service Portal so that users can submit network-related requests.	—
FR-02	Request Information Capture	The catalog item shall capture the required information provided by the requester, including requester details, mobile number, connection type, address, existing connection information where applicable, and other configured request details.	FR-01
FR-03	Mandatory Field Validation	The system shall ensure that all required fields are completed before the requester can submit the network request.	FR-02
FR-04	Conditional Field Handling	The Existing ID field shall be displayed only when the requester selects Existing as the connection type.	FR-02

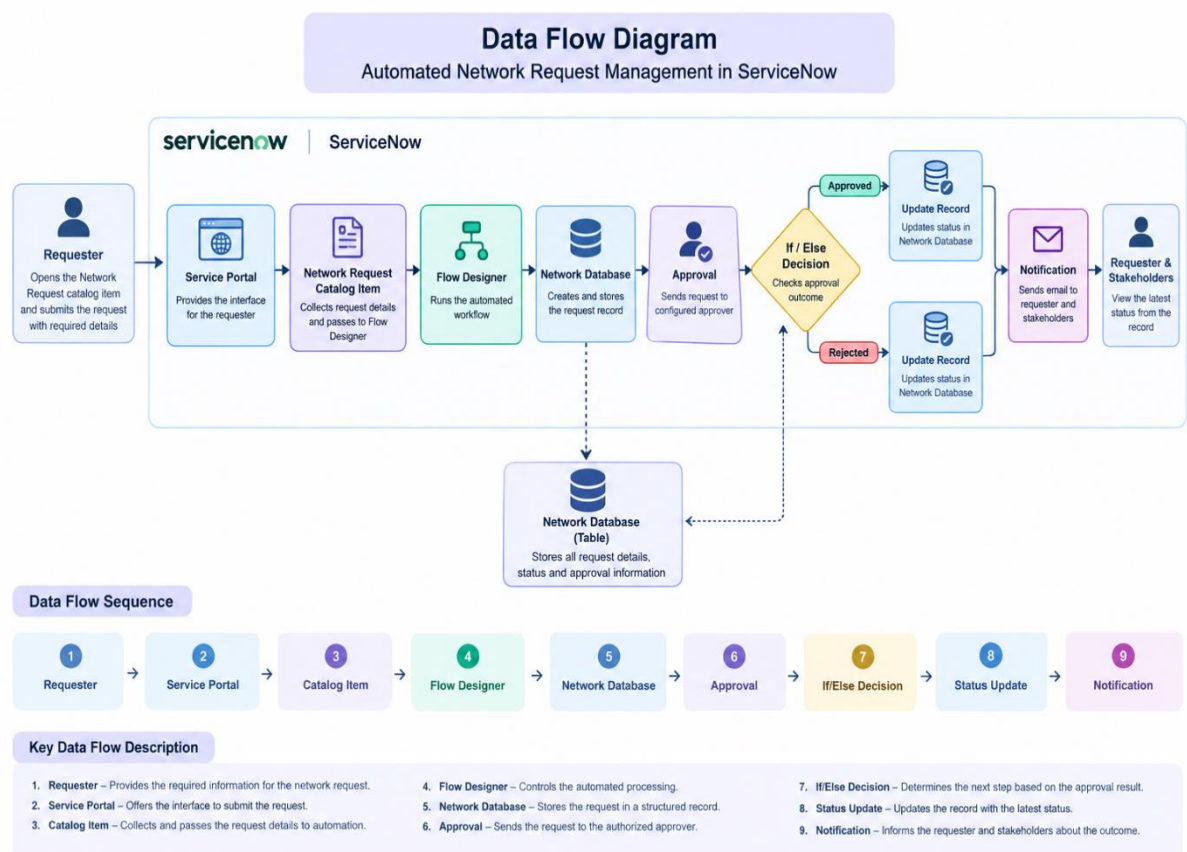
FR-05	Automated Workflow Trigger	Once the Network Request is submitted, the configured ServiceNow Workflow/Flow Designer automation shall start automatically without requiring manual intervention.	FR-01 – FR-04
FR-06	Retrieve Request Details	The workflow shall retrieve the information submitted through the Network Request Catalog Item and use it for further processing.	FR-05
FR-07	Network Database Record Creation	The workflow shall automatically create a corresponding record in the Network Database using the submitted request information.	FR-06
FR-08	Request Data Mapping	The information entered by the requester shall be mapped correctly to the corresponding fields in the Network Database record.	FR-07
FR-09	Approval Request	After the Network Database record is created, the workflow shall generate an approval request for the configured approver.	FR-07
FR-10	Approval Decision	The approver shall be able to review the request and select an appropriate decision such as Approve or Reject .	FR-09
FR-11	Decision-Based Processing	The workflow shall use an If/Else decision to determine the next action based on the approval outcome.	FR-10
FR-12	Record Status Update	The Network Database record shall be updated automatically according to the approval decision.	FR-11
FR-13	Notification	The workflow shall send an email notification to the configured recipients after the request reaches the notification step.	FR-12
FR-14	Request Status Tracking	Users and authorised stakeholders shall be able to view the current status and approval information from the corresponding Network Database record.	FR-12
FR-15	Access Control	Access to the Network Database and related records shall be restricted to authorised ServiceNow users and roles.	FR-07
FR-16	Workflow Verification	The complete request lifecycle shall be tested to verify catalog submission, workflow execution, record creation, approval, decision handling, status update, and notification.	FR-01 – FR-15

Non-Functional Requirements

Quality Attribute	How the Solution Addresses It
Usability	The Network Request is provided through the ServiceNow Service Portal with a structured form, mandatory fields, and conditional field behaviour to make submission simple for the requester.
Automation	The request lifecycle is automated using ServiceNow Workflow/Flow Designer, reducing the need for manual processing after submission.
Security	Access to network request records is restricted through ServiceNow roles and access controls so that only authorised users can access or modify the records.
Reliability	The workflow is designed to consistently execute the configured

	sequence of record creation, approval, decision handling, status update, and notification.
Data Accuracy	Request information submitted through the catalog item is mapped to the Network Database record to maintain consistent and structured request data.
Performance	Automation reduces manual processing time and provides a faster request-handling process. No specific numerical response-time target was defined for this project.
Maintainability	The solution uses ServiceNow's standard catalog, database, workflow/flow, approval, notification, and access-control features, making the configuration easier to maintain.
Scalability	Additional network request fields, approval conditions, notifications, or workflow steps can be incorporated into the existing ServiceNow solution as requirements grow.
Availability	The solution operates within the ServiceNow platform and uses the platform's native services for catalog requests, workflow automation, records, approvals, and notifications.
Traceability	The request information, database record, approval outcome, and updated status are maintained within ServiceNow, allowing the request lifecycle to be tracked.

3.3 Data Flow Diagram



3.4 Technology Stack

Layer-by-Layer Breakdown

Layer	Purpose	Technology Used
User / Client Layer	Provides the interface through which the requester accesses and submits the network request.	ServiceNow Service Portal
Service Catalog Layer	Provides the Network Request catalog item and collects the information required from the requester.	ServiceNow Service Catalog
Form Validation Layer	Handles mandatory fields and controls field behaviour based on the values entered by the requester.	Catalog Configuration, UI Policies / Client-side configuration
Automation Layer	Processes the submitted request automatically, creates the database record, sends the approval request, handles the decision, and updates the record.	ServiceNow Workflow / Flow Designer
Data Layer	Stores the submitted network request information and maintains the request status and approval information.	Custom Network Database Table
Approval Layer	Sends the created network request for review and captures the approver's decision.	ServiceNow Approval
Decision Layer	Determines the next step based on whether the request is approved or rejected.	If/Else workflow logic
Notification Layer	Sends email notifications to the configured users or stakeholders during the request process.	ServiceNow Email / Workflow Notification
Security Layer	Controls access to the network request records based on authorised users and roles.	ServiceNow Roles and ACLs
Runtime Environment	Provides the environment in which the application is configured, tested, and demonstrated.	ServiceNow Personal Developer Instance (PDI)
External Integration Layer	No external application or third-party system is required for the implemented solution.	Not applicable
AI / Intelligence Layer	No AI, machine-learning, or predictive functionality is included in the current project.	Not applicable

Automated Network Request Management – Request Data Flow

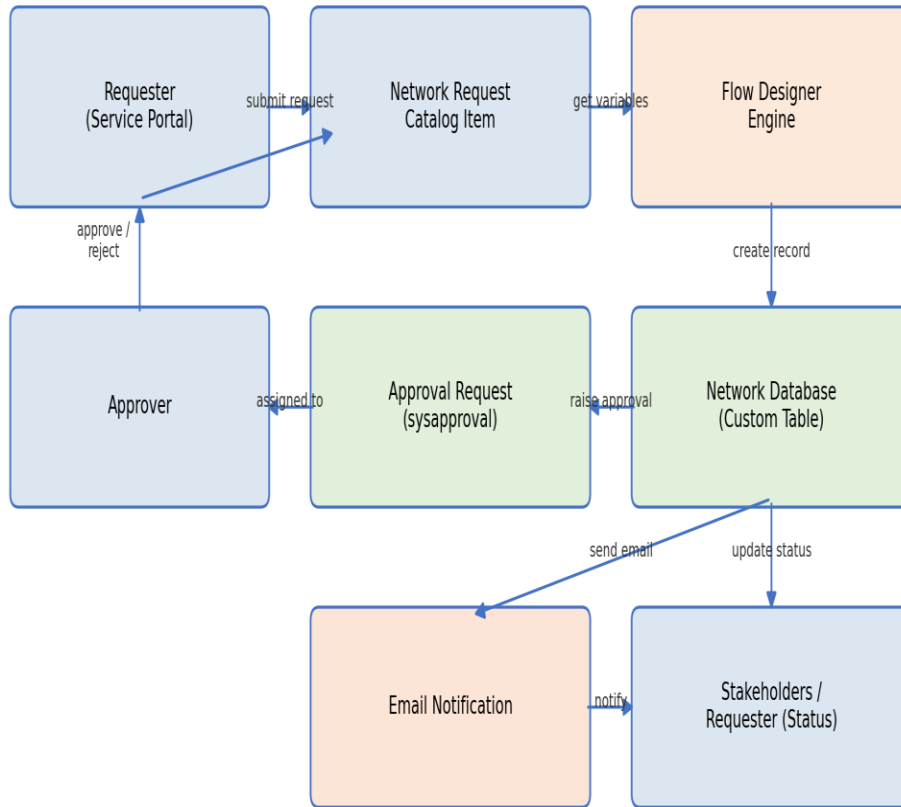


Figure 1: Request flow across the ServiceNow components listed above.

Component Interaction

Service Portal

The Service Portal acts as the entry point for the requester. The requester accesses the Network Request catalog item and provides the required information.

Network Request Catalog Item

The catalog item collects the network request information entered by the requester. Mandatory fields and configured form behaviour help ensure that the required information is provided before submission.

Workflow Automation

After submission, the configured ServiceNow workflow automation processes the request. It retrieves the submitted information, creates the corresponding Network Database record, initiates the approval process, evaluates the approval result, and performs the required record updates.

Network Database

The custom Network Database table stores the request information in a structured ServiceNow record. It also maintains information related to the request status and approval outcome.

Approval

The approval step sends the network request to the configured approver. The approver reviews the request and provides an approval or rejection decision.

If/Else Decision

The workflow evaluates the approval result using conditional logic. The appropriate branch is executed depending on whether the request is approved or rejected.

Record Update

After the approval decision, the Network Database record is automatically updated so that the latest status is available for tracking.

Notification

The configured notification step sends an email to the relevant recipients, keeping them informed about the request process or outcome.

4. PROJECT DESIGN

4.1 Problem Solution Fit

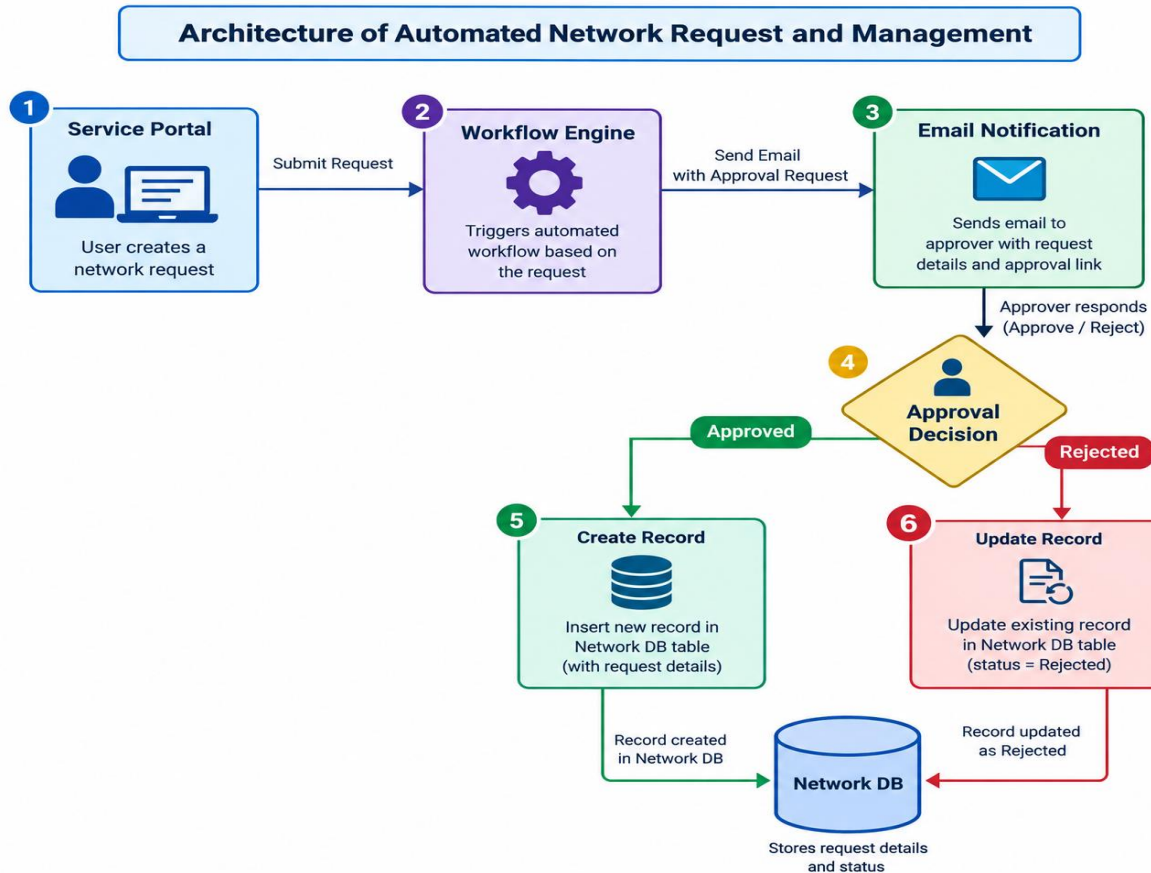
Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS - Employees requesting network resources - IT Help Desk Staff - Network Administrators - System Administrators (Approvers) - Infrastructure & Network Teams - Corporate IT Department Users who need to request, approve and manage network resource requests.	6. CUSTOMER CONSTRAINTS CC - Limited IT resources - Manual approval dependency - Incomplete request details - Approval delays - Data entry mistakes - Lack of centralized database - Communication gaps Challenges and limitations customers face in current manual process.	5. AVAILABLE SOLUTIONS AS - Email-based request and approval - Excel spreadsheets - Manual ticket tracking - Phone communication - Paper request forms - Basic ITSM ticket systems Limitations: - Human errors, slow process, no automation - Poor visibility and reporting	Explore AS, differentiate	
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Submit network resource requests quickly - Avoid manual paperwork and email approvals - Track request status in real time - Ensure accurate request information - Reduce provisioning delays - Automate approval workflow - Store approved requests automatically Tasks customers want to complete and problems they face in current process.	9. PROBLEM ROOT CAUSE RC - Manual request handling - Email-based approvals - No standardized request form - Manual data entry - Lack of workflow automation - No centralized request database - Delayed communication between users and administrators Underlying reasons why the problems exist.	7. BEHAVIOUR BE Current Behaviour: - Send email requests - Wait for administrator response - Frequently follow up for updates - Contact IT through calls/messages - Re-submit forgotten requests - Maintain separate records manually Desired Behaviour: - Submit requests through Service Catalog - Track progress online - Receive automatic notifications - Trust automated workflows	Focus on J&P, tap into BE, understand RC	
Identify strong TR & EM	3. TRIGGERS TR - Employee needs a new network connection - New device installation - Office relocation - Department expansion - Hardware replacement - Existing network issue - Requirement for additional network access	10. YOUR SOLUTION SL Develop a ServiceNow-based Network Resource Request Automation System that: - Uses Service Catalog for request submission - Collects: device_type, connection_type, address, user_name - Automatically triggers Flow Designer - Sends assignment notification to System Administrator - Requests approval from System Administrator - If approved → Inserts data into u_network_database and sends Approval Email (Emerald Green) - If rejected → Sends Rejection Email (Crimson Red) - Provides automated, centralized and traceable request management	8. CHANNELS OF BEHAVIOUR CH 8.1 ONLINE - ServiceNow Service Portal - Corporate Intranet - Email Notifications - Service Catalog - Self-Service Portal 8.2 OFFLINE - IT Help Desk - Phone Support - Network Administration Office - Internal Meetings	Explore online & offline CH of BE	
	4. EMOTIONS: BEFORE / AFTER EM <table><tr><td>Before</td><td>After</td></tr><tr><td> - Frustration due to manual approvals - Long waiting periods - Uncertainty about request status - Duplicate submissions - Communication delays </td><td> - Faster approvals - Transparent tracking - Increased confidence - Reduced workload - Better user satisfaction </td></tr></table>		Before	After	- Frustration due to manual approvals - Long waiting periods - Uncertainty about request status - Duplicate submissions - Communication delays
Before	After				
- Frustration due to manual approvals - Long waiting periods - Uncertainty about request status - Duplicate submissions - Communication delays	- Faster approvals - Transparent tracking - Increased confidence - Reduced workload - Better user satisfaction				

4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Organizations often handle network-related requests manually through emails, calls, or messages. This can lead to delays, missing information, difficulty in tracking requests, and lack of proper approval and status visibility. A structured system is needed to automate and manage network requests efficiently.

2	Idea / Solution description	The proposed solution is an Automated Network Request Management System using ServiceNow . A user submits a network request through the ServiceNow Service Catalog by providing the required details. The request is automatically validated and routed through a workflow for approval. Once approved, the request is processed and its status is updated automatically, providing better tracking and visibility throughout the process.
3	Novelty / Uniqueness	The solution combines ServiceNow Service Catalog, Workflow/Flow Designer, approvals, validations, and automated status updates into a single system. It reduces manual coordination and provides a standardized process for handling network requests. Users can easily submit and track requests without depending on emails or manual follow-ups.
4	Social Impact / Customer Satisfaction	The system improves user satisfaction by making network request submission faster, simpler, and more transparent. Users receive a clear status of their requests and do not need to repeatedly contact support teams for updates. It also helps IT teams reduce repetitive manual work and respond to requests more efficiently.
5	Business Model (Revenue Model)	The solution can be implemented as part of an organization's ServiceNow IT service management platform . Organizations can use it to reduce operational costs, improve IT service efficiency, and minimize manual effort. ServiceNow implementation, customization, maintenance, and support can generate revenue for IT service providers and implementation partners.
6	Scalability of the Solution	The solution can be easily extended to support different types of network requests, such as VPN access, firewall access, IP requests, network configuration, and connectivity issues . Additional workflows, approval levels, notifications, dashboards, and automation can be added as the organization grows. The same approach can also be adapted for other IT service request processes.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was planned in multiple phases to ensure that the ServiceNow application was developed and tested systematically.

Phase	Activities
Phase 1 – Requirement Analysis	Identify the network request problem, users, requirements, and expected workflow.
Phase 2 – Application Design	Design the request form, database structure, workflow, approval process, and security requirements.
Phase 3 – ServiceNow Configuration	Create the application components, catalog item, variables, variable set, and Network Database.
Phase 4 – Workflow	Configure Flow Designer, approval processing, notifications,

Automation	and status updates.
Phase 5 – Security Configuration	Configure roles and ACLs for controlled access.
Phase 6 – Testing	Perform functional testing of request submission, validation, workflow, approval, notifications, status tracking, and access control.
Phase 7 – Final Validation	Perform end-to-end validation and verify that the complete workflow operates correctly.

Sprint Planning

The work can be organized into the following sprints:

Sprint	Main Activities
Sprint 1	Requirement analysis, application planning, and Network Request form design
Sprint 2	Catalog item, variables, variable set, and Network Database configuration
Sprint 3	Flow Designer, approvals, notifications, and status tracking
Sprint 4	ACL configuration, testing, validation, and documentation

6. Functional and Performance Testing

6.1 Performance Testing

Test Scenarios & Results

Test ID	Scenario	Expected Result	Status
TC-01	Verify access to Network Request Service Portal	The ServiceNow portal opens successfully and the user can locate the Network Request option.	Pass
TC-02	Verify Network Request form configuration	The Network Request form opens correctly and displays all configured fields and variable sets.	Pass
TC-03	Validate mandatory request information	The system accepts the request only after the required requester, connection, payment, amount, and address information has been entered correctly.	Pass
TC-04	Verify conditional field behavior	When Type of Connection is changed to Existing , the Existing ID field becomes visible; it remains hidden for	Pass

Test ID	Scenario	Expected Result	Status
		other connection types.	
TC-05	Validate automated request creation	After submitting valid information, the configured Flow Designer flow executes and creates the corresponding record in the Network Database.	Pass
TC-06	Verify approval workflow	The submitted request is routed through the configured approval process and an approval record is generated for the appropriate approver.	Pass
TC-07	Verify request notification mechanism	Configured email notifications are generated and sent to the intended users when the request reaches the relevant workflow stage.	Pass
TC-08	Verify request lifecycle status	Changes in the request workflow are reflected in the Network Database through the configured Approval Status field.	Pass
TC-09	Verify table-level security	Users can perform only the read/write operations permitted by the configured ACL rules on the Network Database.	Pass
TC-10	Validate complete automated workflow	A valid request successfully progresses from submission through validation, record creation, approval, notification, status tracking, and completion without workflow errors.	Pass

Functional & Performance Testing Results

Test ID	Scenario	Status
TC-01	Verify ServiceNow portal accessibility	PASS
TC-02	Access and verify the Network Request catalog item	PASS
TC-03	Verify entry of required network request details	PASS
TC-04	Validate conditional display of Existing ID field	PASS

TC-05	Verify successful submission and automated request record creation	PASS
TC-06	Validate automatic approval request creation	PASS
TC-07	Verify automated notification for request updates	PASS
TC-08	Verify tracking of request approval and processing status	PASS
TC-09	Validate user access permissions for Network Database records	PASS

9/9 functional scenarios passed

7. Results

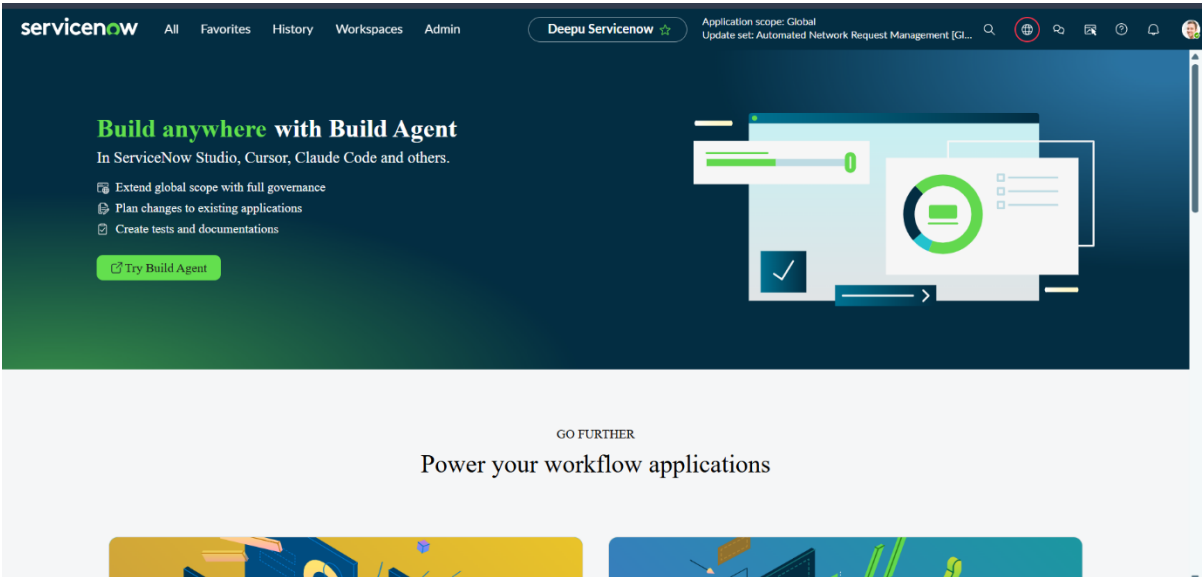


Fig.1 Servicenow Home page

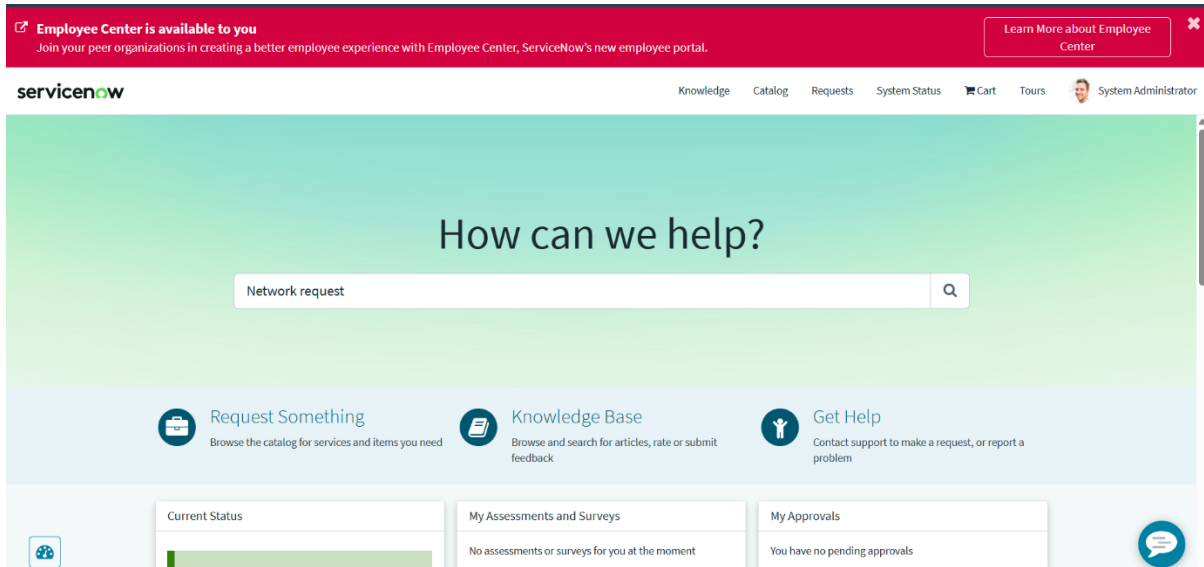


Fig.2 Service Portal

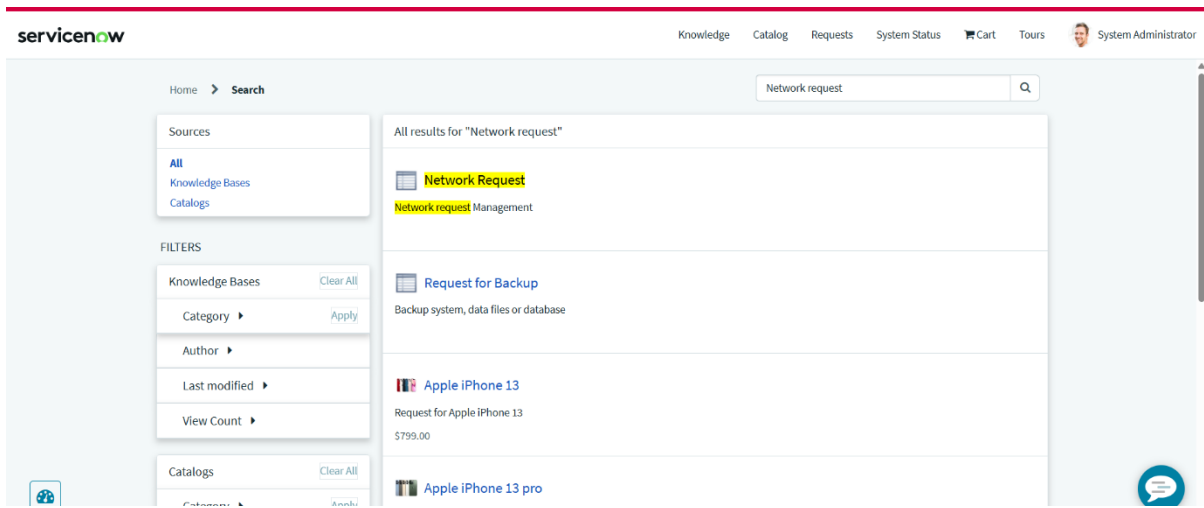


Fig.3 Search and open Network Request

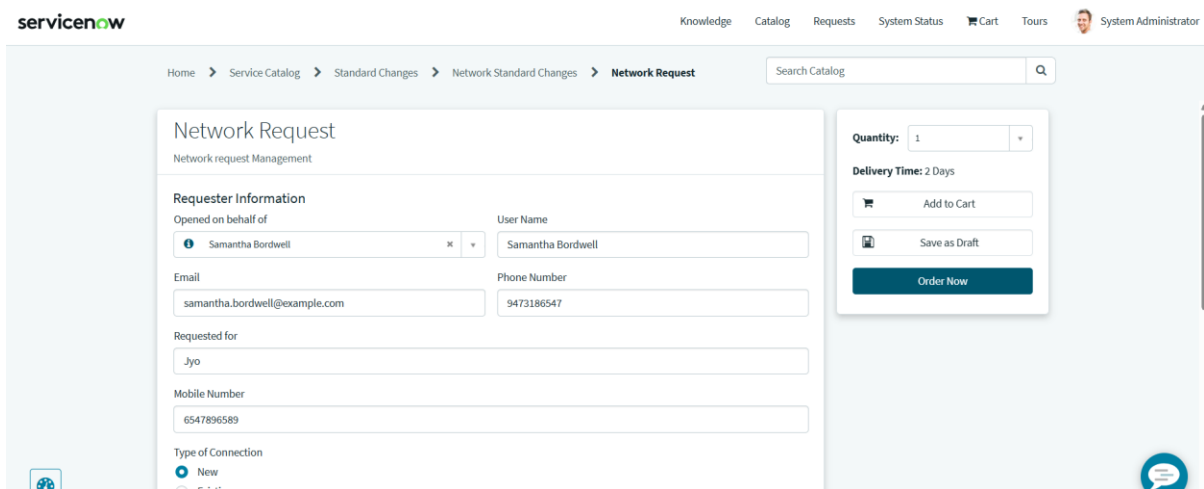


Fig.4 Enter the details and click on order now

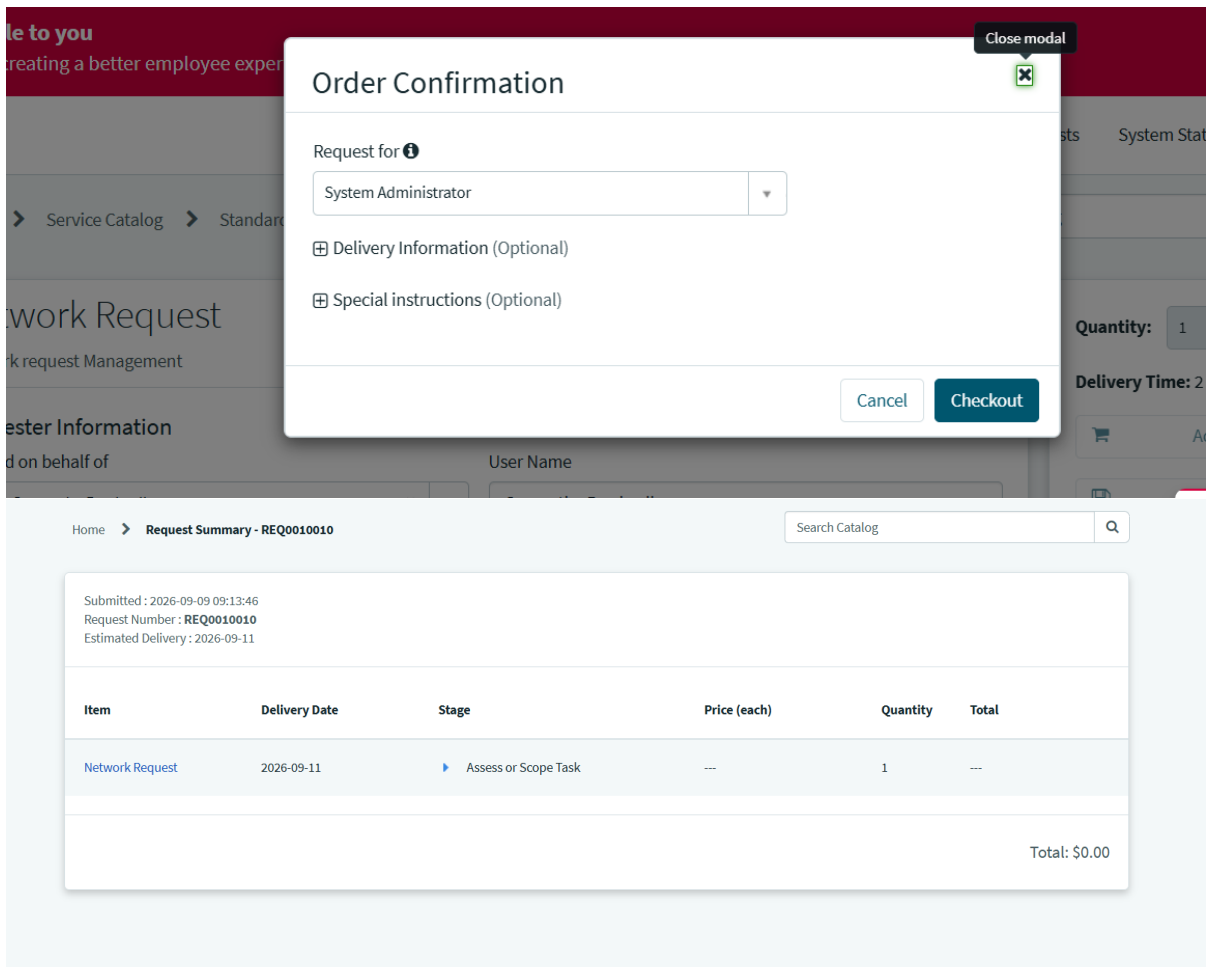


Fig.5 Checkout and Request Summary

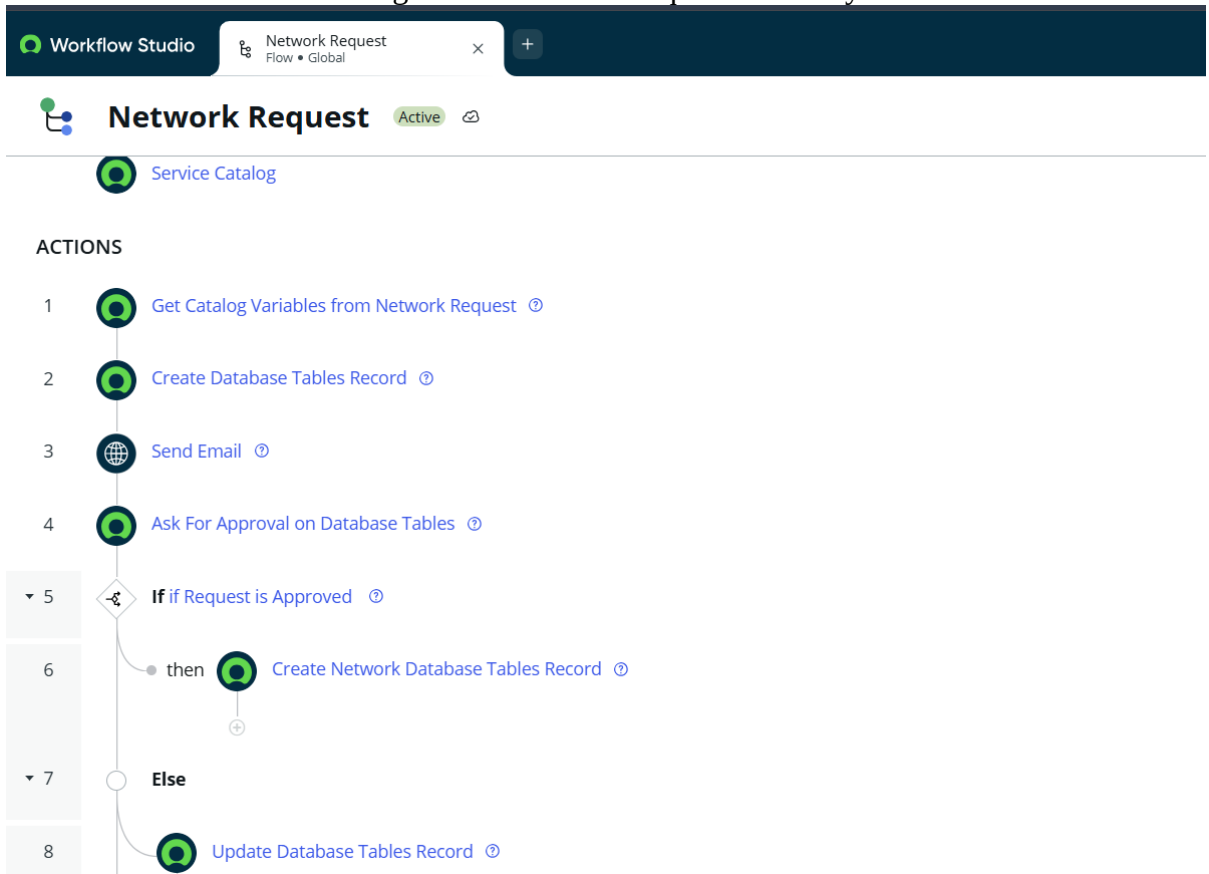


Fig.6 Work Flow for Automation

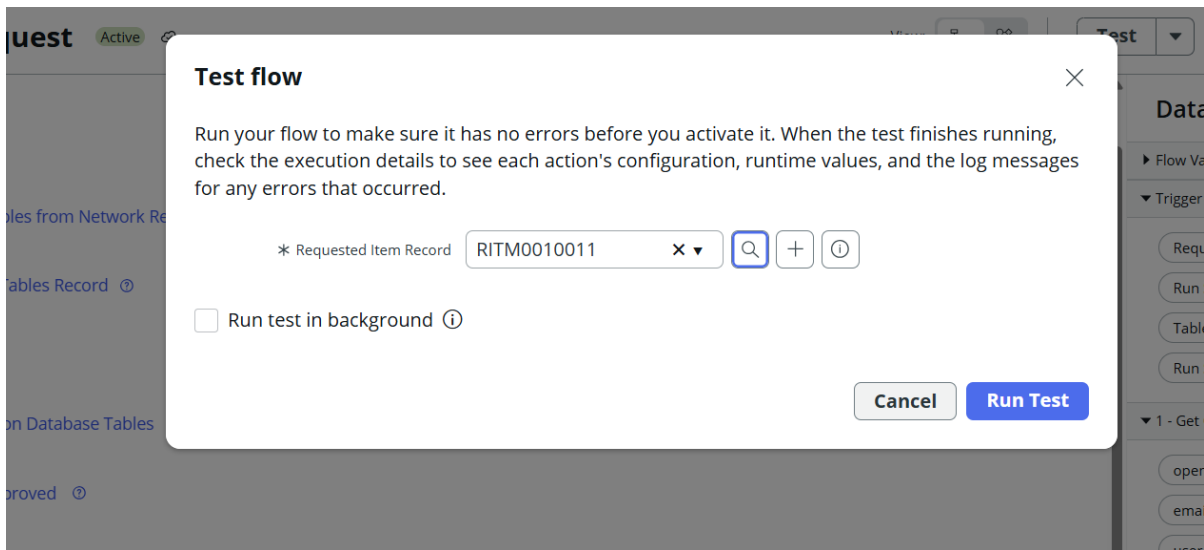


Fig.7 Test the flow by selecting the record

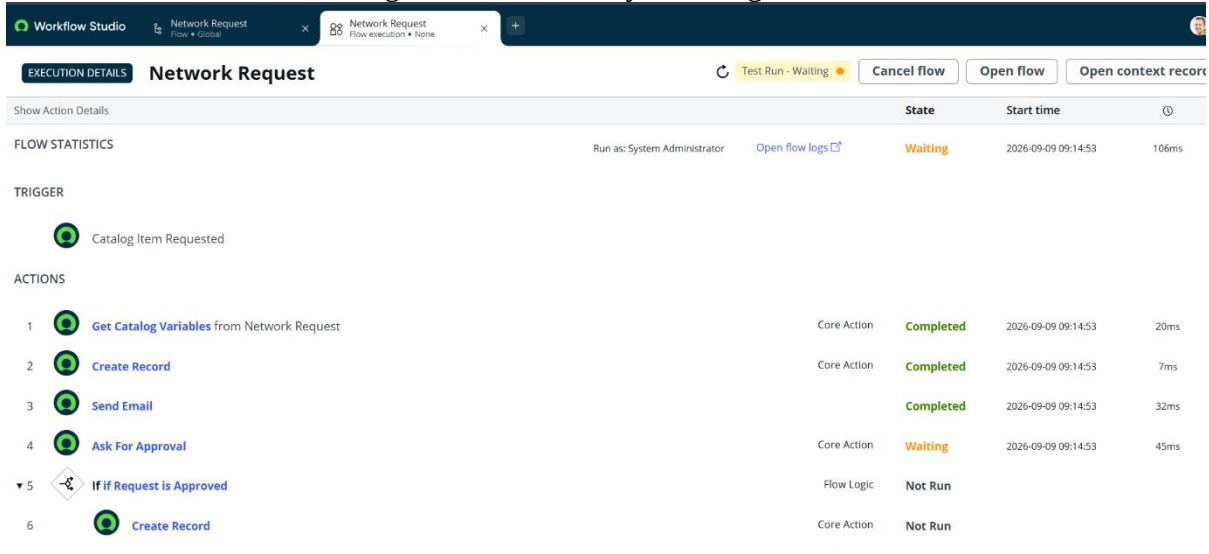


Fig.8 Its waiting for Approval and email sent

	Created	Recipients	Subject	Type	Notification type	User ID
☐	2026-09-09 01:00:04	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
☐	2026-09-09 00:00:04	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
☐	2026-09-09 01:00:04	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
☐	2026-09-09 09:13:54	fred.luddy@example.com	Catalog Task SCTASK0010020 has been assigned to group Software	send-ready	SMTP	(empty)
☐	2026-09-09 09:13:54	admin@example.com	Request REQ0010010 was created	send-ready	SMTP	(empty)
☐	2026-09-09 09:13:54	admin@example.com	Request REQ0010010 was approved	send-ready	SMTP	(empty)
☐	2026-09-09 01:00:04	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
☐	2026-09-09 09:14:53	Prashanth@gmail.com	Database record is created	send-ready	SMTP	(empty)

Fig.9 Email

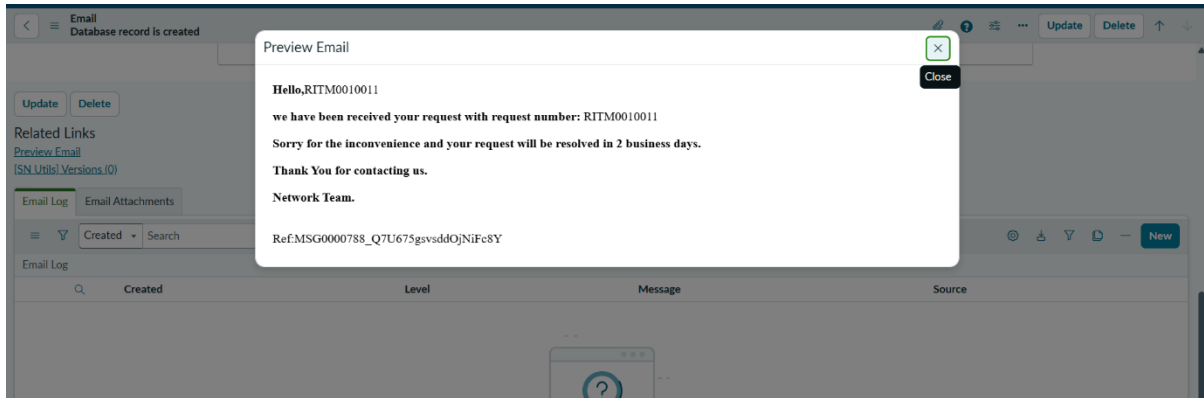


Fig.10 Email Preview

State	Approver	Comments	Approval for	Created
Requested	System Administrator		(empty)	2026-09-09 09:14:53
Rejected	System Administrator	2026-09-08 05:51:43 - System Administrat	(empty)	2026-09-08 05:50:22
Approved	System Administrator		(empty)	2026-09-08 05:46:44
Rejected	System Administrator	2026-09-08 05:34:18 - System Administrat	(empty)	2026-09-08 05:33:25
Approved	System Administrator	2026-09-08 05:32:57 - System Administrat	(empty)	2026-09-08 05:24:11
Approved	System Administrator		(empty)	2026-09-06 00:22:03
Approved	System Administrator		(empty)	2026-09-06 00:19:19
Approved	System Administrator		(empty)	2026-09-06 00:12:48
Approved	System Administrator		(empty)	2026-09-06 00:08:12
Approved	System Administrator		(empty)	2026-09-05 23:36:40

Fig.11 Approval Request

Approver: System Administrator

State: Requested

Approving: Network Database Tables: RITM0010011

Approval Reason:

Comments: Comments

Post

Activities: 1

System Administrator

Field changes • 2026-09-09 09:14:53

Approver: System Administrator

State: Requested

State	Approver	Comments	Approval for	Created
Approved	System Administrator		(empty)	2026-09-09 09:14:53
Rejected	System Administrator	2026-09-08 05:51:43 - System Administrat	(empty)	2026-09-08 05:50:22
Approved	System Administrator		(empty)	2026-09-08 05:46:44
Rejected	System Administrator	2026-09-08 05:34:18 - System Administrat	(empty)	2026-09-08 05:33:25
Approved	System Administrator	2026-09-08 05:32:57 - System Administrat	(empty)	2026-09-08 05:24:11
Approved	System Administrator		(empty)	2026-09-06 00:22:03
Approved	System Administrator		(empty)	2026-09-06 00:19:19
Approved	System Administrator		(empty)	2026-09-06 00:12:48
Approved	System Administrator		(empty)	2026-09-06 00:08:12
Approved	System Administrator		(empty)	2026-09-05 23:36:40
Approved	System Administrator		(empty)	2026-09-05 23:35:58
Approved	System Administrator		(empty)	2026-09-05 23:31:12

Fig.12 Request Approval

Workflow Studio Network Request Flow execution: None

EXECUTION DETAILS Network Request

Test Run - Completed Open flow Open context record

Catalog Item Requested

ACTIONS

Step	Action	Status	Timestamp	Duration
1	Get Catalog Variables from Network Request	Completed	2026-09-09 09:14:53	20ms
2	Create Record	Completed	2026-09-09 09:14:53	7ms
3	Send Email	Completed	2026-09-09 09:14:53	32ms
4	Ask For Approval	Completed	2026-09-09 09:14:53	187ms
5	If if Request is Approved	Evaluated - True	2026-09-09 09:16:32	6ms
6	Create Record	Completed	2026-09-09 09:16:32	6ms
7	Else	Not Run		
8	Update Record	Not Run		

ERROR HANDLER

Fig.13 Retest and check the flow

servicenow All Favorites History Admin Deepu Servicenow - Network Database Tables | Deep...

Application scope: Global Update set: Automated Network Request Management [G...]

Network Database Tables Updated Search

Actions on selected rows... New

Number	Approval Status	Assignment to	Customer Address	Mobile Number	Mode of Payment	Updated	Total Amount	Type of Connection
DAT0001037	Approved	Abraham Lincoln	Sontyam	6547896589	Card	2026-09-09 09:16:32	1650	New
RITM0010011	Requested	(empty)	Sontyam	6547896589	Card	2026-09-09 09:14:53	1650	New
RITM0010010	Rejected	Abraham Lincoln	Yendada	6589632145	Card	2026-09-08 05:51:43	1000	New
DAT0001036	Approved	Abraham Lincoln	Vizag	8569745689	UPI	2026-09-08 05:47:16	3560	New
RITM0010009	Requested	(empty)	Vizag	8569745689	UPI	2026-09-08 05:46:44	3560	New
RITM0010008	Rejected	Abraham Lincoln	Dairy Farm	9632136988	UPI	2026-09-08 05:34:18	1565	New
DAT0001033	Approved	Abraham Lincoln	Dairy Farm	9632136988	UPI	2026-09-08 05:24:38	1565	New
RITM0010008	Requested	(empty)	Dairy Farm	9632136988	UPI	2026-09-08 05:24:06	1565	New
DAT0001032	Approved	Abraham Lincoln	Delhi	6987456985	Card	2026-09-06 00:22:18	100	New
RITM0010007	Requested	(empty)	Delhi	6987456985	Card	2026-09-06 00:22:03	100	New
DAT0001031	Approved	Abraham Lincoln	HYD	7896541235	UPI	2026-09-06 00:19:37	5600	New
RITM0010006	Requested	(empty)	HYD	7896541235	UPI	2026-09-06 00:19:19	5600	New
DAT0001030	Approved	Abraham Lincoln	Vizag	8745963256	Card	2026-09-06 00:13:14	5000	New
RITM0010005	Requested	(empty)	Vizag	8745963256	Card	2026-09-06 00:12:48	5000	New
DAT0001029	Approved	Abraham Lincoln	HYD	7896541235	UPI	2026-09-06 00:08:45	5600	New

Fig.14 Creates a record as Approved

Home > Service Catalog > Standard Changes > Network Standard Changes > Network Request

Search Catalog

Network Request

Network request Management

Requester Information

Opened on behalf of: Zane Sulikowski

User Name: Zane Sulikowski

Email: zane.sulikowski@example.com

Phone Number: 7854967856

Requested for: deepu

Mobile Number: 9632147856

Type of Connection: New

Fig.15 Create record to check what happens if approval rejected

servicenow All Favorites History Admin : Deepu Servicenow - Network Database Tables: RITM... Application scope: Global Update set: Automated Network Request Management [GL...]

Approval Network Database Tables: RITM0010012 [Update] [Approve] [Reject] [Delete]

Approver: System Administrator Approving: Network Database Tables: RITM0010012

State: Rejected

Approval Reason:

* Comments: Do not contain required data

Activities: 1 System Administrator Field changes: 2026-09-09 09:20:42

Update Approve Reject Delete

Summary of item being approved

Network Database Tables

Fig.16 Reject and give comments why it is rejected

servicenow All Favorites History Admin : Deepu Servicenow - Approvals | Deepu Servicenow Application scope: Global Update set: Automated Network Request Management [GL...]

Approvals Created Search Actions on selected rows...

All > Sys ID = NULL .or. Approver = System Administrator

State	Approver	Comments	Approval for	Created
Rejected	System Administrator	2026-09-09 09:21:24 - System Administrator	(empty)	2026-09-09 09:20:42
Approved	System Administrator		(empty)	2026-09-09 09:14:53
Rejected	System Administrator	2026-09-08 05:51:43 - System Administrator	(empty)	2026-09-08 05:50:22
Approved	System Administrator		(empty)	2026-09-08 05:46:44
Rejected	System Administrator	2026-09-08 05:34:18 - System Administrator	(empty)	2026-09-08 05:33:25
Approved	System Administrator	2026-09-08 05:32:57 - System Administrator	(empty)	2026-09-08 05:24:11
Approved	System Administrator		(empty)	2026-09-06 00:22:03
Approved	System Administrator		(empty)	2026-09-06 00:19:19
Approved	System Administrator		(empty)	2026-09-06 00:12:48
Approved	System Administrator		(empty)	2026-09-06 00:08:12
Approved	System Administrator		(empty)	2026-09-05 23:36:40
Approved	System Administrator		(empty)	2026-09-05 23:35:58

Fig.17 Approval Rejected

Workflow Studio Network Request Network Request Network Request

EXECUTION DETAILS Network Request Test Run - Completed Open flow Open context record Refresh Flow Data

ACTIONS	Core Action	Status	Timestamp	Duration
1 Get Catalog Variables from Network Request	Core Action	Completed	2026-09-09 09:20:42	25ms
2 Create Record	Core Action	Completed	2026-09-09 09:20:42	5ms
3 Send Email	Core Action	Completed	2026-09-09 09:20:42	21ms
4 Ask For Approval	Core Action	Completed	2026-09-09 09:20:42	144ms
5 If Request is Approved	Flow Logic	Evaluated - False	2026-09-09 09:21:24	0ms
6 Create Record	Core Action	Not Run		
7 Else	Flow Logic	Completed	2026-09-09 09:21:24	13ms
8 Update Record	Core Action	Completed	2026-09-09 09:21:24	13ms

ERROR HANDLER

Fig.18 Retest after rejected

Number	Approval Status	Assignment to	Customer Address	Mobile Number	Mode of Payment	Updated	Total Amount	Type of Connection
RITM0010012	Rejected	Abraham Lincoln	Carshed	9632147856	Card	2026-09-09 09:21:24	15000	New
DAT0001037	Approved	Abraham Lincoln	Sontyam	6547896589	Card	2026-09-09 09:16:32	1650	New
RITM0010011	Requested	(empty)	Sontyam	6547896589	Card	2026-09-09 09:14:53	1650	New
RITM0010010	Rejected	Abraham Lincoln	Yendada	6589632145	Card	2026-09-08 05:51:43	1000	New
DAT0001036	Approved	Abraham Lincoln	Vizag	8569745689	UPI	2026-09-08 05:47:16	3560	New
RITM0010009	Requested	(empty)	Vizag	8569745689	UPI	2026-09-08 05:46:44	3560	New
RITM0010008	Rejected	Abraham Lincoln	Dairy Farm	9632136988	UPI	2026-09-08 05:34:18	1565	New
DAT0001033	Approved	Abraham Lincoln	Dairy Farm	9632136988	UPI	2026-09-08 05:24:38	1565	New
RITM0010008	Requested	(empty)	Dairy Farm	9632136988	UPI	2026-09-08 05:24:06	1565	New
DAT0001032	Approved	Abraham Lincoln	Delhi	6987456985	Card	2026-09-06 00:22:18	100	New
RITM0010007	Requested	(empty)	Delhi	6987456985	Card	2026-09-06 00:22:03	100	New

Fig.19 Updates a record that its rejected

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual processing of network requests.
- Provides a centralized request management system.
- Simplifies network request submission.
- Provides structured and validated request information.
- Supports dynamic form behavior.
- Automates approval processing.
- Generates automated notifications.
- Provides request and approval status tracking.
- Improves visibility of the request lifecycle.
- Provides role- and ACL-based access control.
- Reduces repeated communication between users and administrators.
- Provides an organized and traceable workflow.

Disadvantages

- Requires a configured ServiceNow environment.
- Initial configuration and workflow development require ServiceNow knowledge.
- Changes to the request process may require administrator configuration.
- The system depends on the availability of the ServiceNow platform.
- Advanced customization may require additional scripting or platform configuration.
- Performance at very large scale would require separate load and scalability testing.

9. CONCLUSION

The **Automated Network Request Management in ServiceNow** project provides a centralized and automated approach for managing network-related service requests.

The system enables users to submit requests through the ServiceNow Service Portal using a structured Network Request Catalog Item. The configured form collects the required information and supports conditional field behavior.

Flow Designer automates the request processing, approval, notification, and status management activities. Network request information is maintained in a centralized Network Database, while ACL configurations help control access to records.

Functional testing confirmed that all defined test scenarios passed successfully. Overall, the project demonstrates how ServiceNow can be used to automate and manage a complete service request lifecycle efficiently.

10. FUTURE SCOPE

The system can be enhanced further with the following features:

- Integration with external network management systems.
- Automated assignment of requests to specific network teams.
- SLA configuration for monitoring request resolution time.
- Escalation notifications for delayed approvals or requests.
- Dashboard for administrators to monitor network requests.
- Reports based on request type, status, and processing time.
- Mobile access for request monitoring.
- Advanced analytics for identifying request trends.
- Integration with additional enterprise services.
- Automated request closure after successful fulfillment.
- Enhanced audit and monitoring capabilities.
- Self-service knowledge articles for common network issues.

11. APPENDIX

Source Code (if any)

The project is primarily developed and configured using the ServiceNow platform. The implementation consists mainly of ServiceNow configuration components such as:

- Service Catalog configuration.
- Catalog variables.
- Variable sets.
- ServiceNow tables.

- Flow Designer workflows.
- Approval configuration.
- Notification configuration.
- ACL/security configuration.
- ServiceNow scripting/configuration where required.

If any custom scripts were used during implementation, they can be included in this section.

Dataset Link

This project does not require a separate external dataset. Network request information is collected through the ServiceNow request form and maintained within the configured ServiceNow tables.

Dataset: Not Applicable / ServiceNow internal request data.

GitHub & Project Demo Link

GitHub Repository:

<https://github.com/VijjapuMadhuri5/Automated-Network-Request-Management-in-ServiceNow>

Project Demo:

https://drive.google.com/file/d/1pfB9gtfKB5DrPiPJamHhgulWbREvX4T0/view?usp=drive_link